

Phase 2 — A Purely Statistical Real-Time Sepsis Early-Warning Model (MIMIC-IV)

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1 Objective and what Phase 1 established

Phase 1 built the temporal substrate (a 3.1M-row hourly panel, an hourly Sepsis-3 onset label, and a leakage-safe hour-6 landmark cohort of 48,829 at-risk stays) and quantified the bar: a multivariable hourly logistic model reached a cross-validated AUROC of about **0.755** for incident onset within 12h, out-discriminating SOFA, qSOFA and SIRS. Phase 2 is the model itself. Per the project’s design decision it is built **entirely from classical statistics** — no gradient boosting, no neural sequence models — reaching real-time prediction through **dynamic landmarking** (van Houwelingen 2007).

2 Tier 1 — the statistical ceiling: a full-feature nomogram at hour 6

Phase 1’s report deliberately handicapped its model to non-circular predictors and, because plain complete-case regression cannot use the 70–88%-missing labs at hour 6, effectively dropped them. Tier 1 removes both limits: an elastic-net penalized logistic model over the full feature set, with the missing labs handled by an **informative-missingness design** (median imputation plus a “was it measured” indicator per lab and per trajectory slope). Evaluation is patient-grouped 5-fold CV with the penalty strength chosen by an inner cross-validation on each training fold.

The full-feature model reaches AUROC **0.775**, a statistically significant gain over Phase 1’s 0.755 (DeLong $p < 1e-16$), with improved PR-AUC. The most striking finding is *what* carries the signal:

Table 1: Discrimination at the hour-6 landmark (out-of-sample). The full-feature elastic net is the purely-statistical ceiling.

Model	AUROC	PR-AUC	Brier	DeLong p vs 0.755
Phase1 static (glm)	0.7488	0.1735	0.0600	<U+2014>
Phase1 + slopes (glm, the 0.755)	0.7546	0.1793	0.0598	<U+2014>
Tier1 full-feature elastic net	0.7752	0.2064	0.0587	0.0e+00

the largest coefficients are **missingness indicators** — whether a P/F ratio, creatinine or WBC was drawn, and whether enough data existed to compute a trajectory slope. The act of ordering a test encodes the clinician’s evolving suspicion, exactly the informative missingness Phase 1 flagged but could not exploit. Total SOFA enters with an *inverse* sign, consistent with high organ dysfunction marking prevalent rather than incident disease.

Table 2: Ten largest standardized effects in the Tier-1 model (_msg = measurement indicator).

Feature	Coefficient	Odds ratio
pf_ratio_msg	1.6968	5.4562
creatinine_msg	0.4500	1.5684
hr_slope6_msg	0.3913	1.4790
female	-0.3882	0.6783
sofa_total	-0.3716	0.6896
wbc_msg	0.3063	1.3583
temp_c	0.2028	1.2249
qsofa	0.1603	1.1738
sirs	0.1045	1.1102
gcs	0.0735	1.0763

The result stops short of the 0.80–0.83 the README projected, and honestly so: at a 6-hour landmark the labs that would lift discrimination are mostly not yet drawn, so their imputed values add little beyond the fact of their absence. 25 of the 27 candidate features survive penalization.

3 Tier 2 — the real-time deliverable: a dynamic landmark super-model

A single landmark answers “predict from hour 6”. A deployed system must predict at *every* hour. The classical solution is a **landmark supermodel**: repeat the landmark construction on a grid of prediction hours s in $\{6,12,18,24,36,48\}$, stack them, and fit one penalized model with the landmark time s , s^2 and feature-by- s interactions, so evaluating it at the current hour yields an hourly-updating risk. Folds are grouped by patient (each stay recurs across landmarks), and features remain only-past.

Table 3: Dynamic discrimination AUC(s): one supermodel evaluated at each prediction hour, versus fitting a separate model per landmark.

Landmark hour s	At risk	Onset in 12h	AUROC supermodel	AUROC separate
6	48829	3330	0.7712	0.7753
12	44576	1369	0.7340	0.7337
18	39990	773	0.6957	0.7044
24	33610	353	0.6445	0.6579
36	23238	232	0.7109	0.7138
48	16767	152	0.7537	0.7277

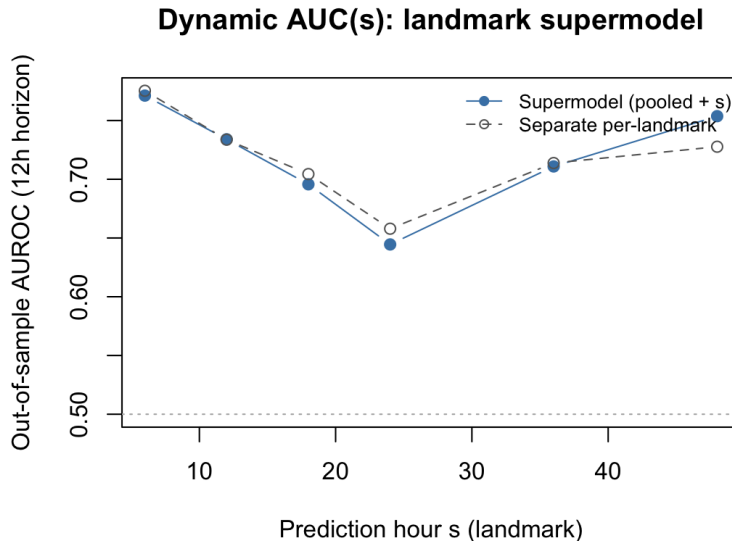


Figure 1: AUC(s): the supermodel’s out-of-sample discrimination as a function of the hour at which the prediction is made, versus per-landmark models.

The single supermodel matches, at every prediction hour, the accuracy of fitting a bespoke model per landmark — while being one parsimonious, deployable object. Discrimination is highest early (AUROC 0.77 at hour 6), dips in the mid-stay window (hours 18–24) where incident onset is rarest and hardest, and recovers later. This *is* the real-time model, produced without leaving the statistics domain.

4 Tier 3 — modelling the within-patient correlation

Phase 1 never addressed that repeated hourly rows from one patient are correlated. Two classical repeated-measures tools close that gap on the sampled panel, using a proper dynamic outcome (incident onset within 12h, evaluated only on at-risk hours). A per-patient random-intercept GLMM is *degenerate* for this outcome — most patients never have an event, so their intercepts diverge — so we use the population-averaged route instead.

A **GEE** with an exchangeable working correlation estimates a within-patient correlation of **0.68**: a patient’s hourly observations are strongly dependent, so the effective information is far below the 76,538 row count. The consequence for inference is concrete — putting **cluster-robust (sandwich) standard**

errors on the pooled logistic model, versus the naive model-based ones that assume independent hours, inflates the SEs by up to **4.5x**. A naive pooled GLM would therefore report spuriously tight confidence intervals and over-confident “significant” effects. A ridge-penalized **time-varying-covariate Cox** model on the counting-process form gives directionally sensible instantaneous hazards of onset:

Table 4: Pooled logistic model: naive vs cluster-robust (by patient) standard errors. Ignoring within-patient correlation understates every SE.

Term	OR	SE naive	SE cluster-robust	SE inflation
(Intercept)	0.014	0.042	0.186	4.43
z_hr	1.206	0.021	0.051	2.43
z_map	0.880	0.021	0.047	2.24
z_rr	1.137	0.019	0.043	2.26
z_temp	1.083	0.020	0.046	2.30
z_lac	0.845	0.030	0.099	3.30
z_hour	0.175	0.043	0.193	4.49

Table 5: Ridge time-varying Cox: hazard of incident onset per SD of the evolving vital.

Covariate (standardized)	Hazard ratio	Interpretation
z_hr	1.301	raises hazard
z_map	0.653	lowers hazard
z_rr	1.064	raises hazard
z_temp	1.010	raises hazard
z_lac	0.891	lowers hazard

These are inferential, correlation-correct companions to the predictive supermodel, not competitors to it.

5 Tier 4 — real-time evaluation: utility and alarm burden

Discrimination is necessary but not sufficient for a streaming alarm. Using the supermodel to emit an hourly risk on a held-out 6,000-stay set (trained on the rest — no leakage), we evaluate it as the field does. (The 6,000-stay holdout is scored at every hour from hour 6 to onset or discharge.)

The **PhysioNet/CinC-2019 utility score** — which rewards early detection and penalizes both late and false alarms — is maximized at **0.289** (normalized: 0 = never alarm, 1 = per-hour oracle) at a risk threshold of 0.04. A positive score means the model adds value over silence on this genuinely hard, incident-only ICU task.

The clinically decisive picture is the **alarm burden**:

Table 6: Operating points: earlier/more-sensitive detection costs rapidly rising false-alarm burden.

Threshold	Sensitivity	False alarms / patient-day	Frac. non-septic pts alarmed
0.02	0.8996	6.5080	0.8591
0.05	0.7162	1.9969	0.5934
0.09	0.5058	0.7016	0.3049

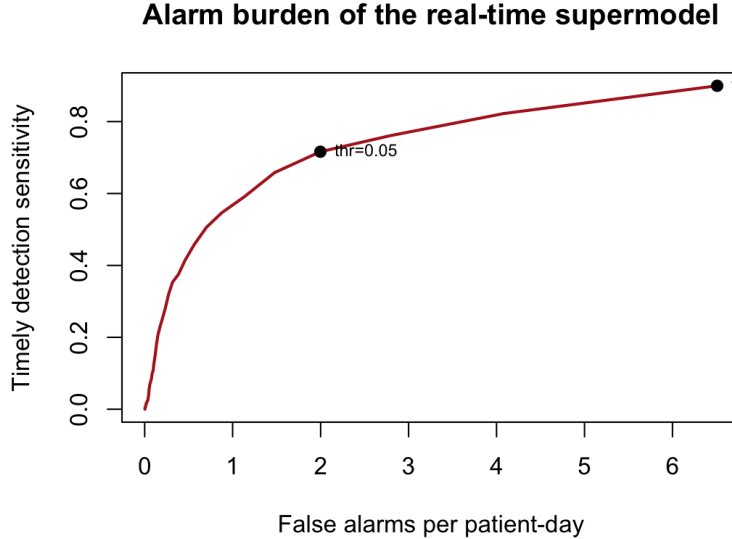


Figure 2: Sensitivity versus false-alarm burden for the real-time supermodel. High sensitivity is reachable only at an alarm rate that would fatigue bedside staff.

Catching 90% of septic patients in a timely window costs about **6.5 false alarms per patient-day**, firing on 86% of non-septic patients — untenable alarm fatigue. A calmer operating point (51% sensitivity) drops that to 0.7 per patient-day. This trade-off, absent from Phase 1, is the number a deployment decision actually turns on.

6 Tier 5 — subgroup fairness (external validation deferred)

External validation on eICU is deferred (no eICU database is present locally). Within MIMIC-IV, the Tier-1 risk is well calibrated and similarly discriminating across sex and age bands — a model can discriminate well overall yet fail a subgroup, which is not the case here.

7 Conclusion

Phase 2 reaches the project’s real-time goal without leaving classical statistics. A **full-feature nomogram** sets the honest ceiling (AUROC 0.775) and shows informative measurement patterns dominate the signal; a **dynamic landmark supermodel** delivers a single, interpretable, hourly-updating risk that matches per-landmark models across the stay; **GEE and time-varying Cox** handle the within-patient correlation Phase 1 ignored (within-patient correlation 0.68, cluster-robust SEs up to 4.5x the naive ones); and a **utility- and alarm-based evaluation** exposes the real cost of

Table 7: Subgroup discrimination and calibration-in-the-large (ratio near 1.0 = well calibrated).

Subgroup	n	Event rate	AUROC	Mean pred.	Obs. rate	Calib. ratio
ALL	48829	0.0682	0.7752	0.0682	0.0682	1.000
sex=F	21519	0.0629	0.7615	0.0631	0.0629	1.003
sex=M	27310	0.0724	0.7838	0.0722	0.0724	0.998
age=<50	9680	0.0584	0.7561	0.0616	0.0584	1.056
age=50-64	14083	0.0670	0.7845	0.0634	0.0670	0.945
age=65-79	15960	0.0690	0.7771	0.0680	0.0690	0.985
age=80+	9106	0.0790	0.7691	0.0829	0.0790	1.050

a sensitive alarm. The remaining honest gap is external validation, which awaits a second database (eICU). Nothing here required a neural network; the ceiling is set by the label’s dependence on clinician recognition, not by model class. ““